Software Design Document (SDD)
AI-Powered Intelligent Interview Question Management System

Version: 1.0

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1. Introduction

1.1 Purpose

The purpose of this Software Design Document (SDD) is to describe the technical design and architecture of the AI-Powered Intelligent Interview Question Management System. It explains how the system is structured, how its components interact, and how functional requirements defined in the Software Requirements Specification (SRS) are implemented. This document serves as the technical blueprint for implementing the proposed system.

1.2 Scope

This document covers the overall system architecture, application workflow, component design, and API design of the AI-Powered Intelligent Interview Question Management System. The system enables authorized users to securely manage interview questions, perform AI-powered semantic searches using Retrieval-Augmented Generation (RAG), apply advanced filters, and retrieve context-aware responses. It also defines the interaction between the frontend, backend, AI services, and data storage components.

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2. System Overview

Project Overview

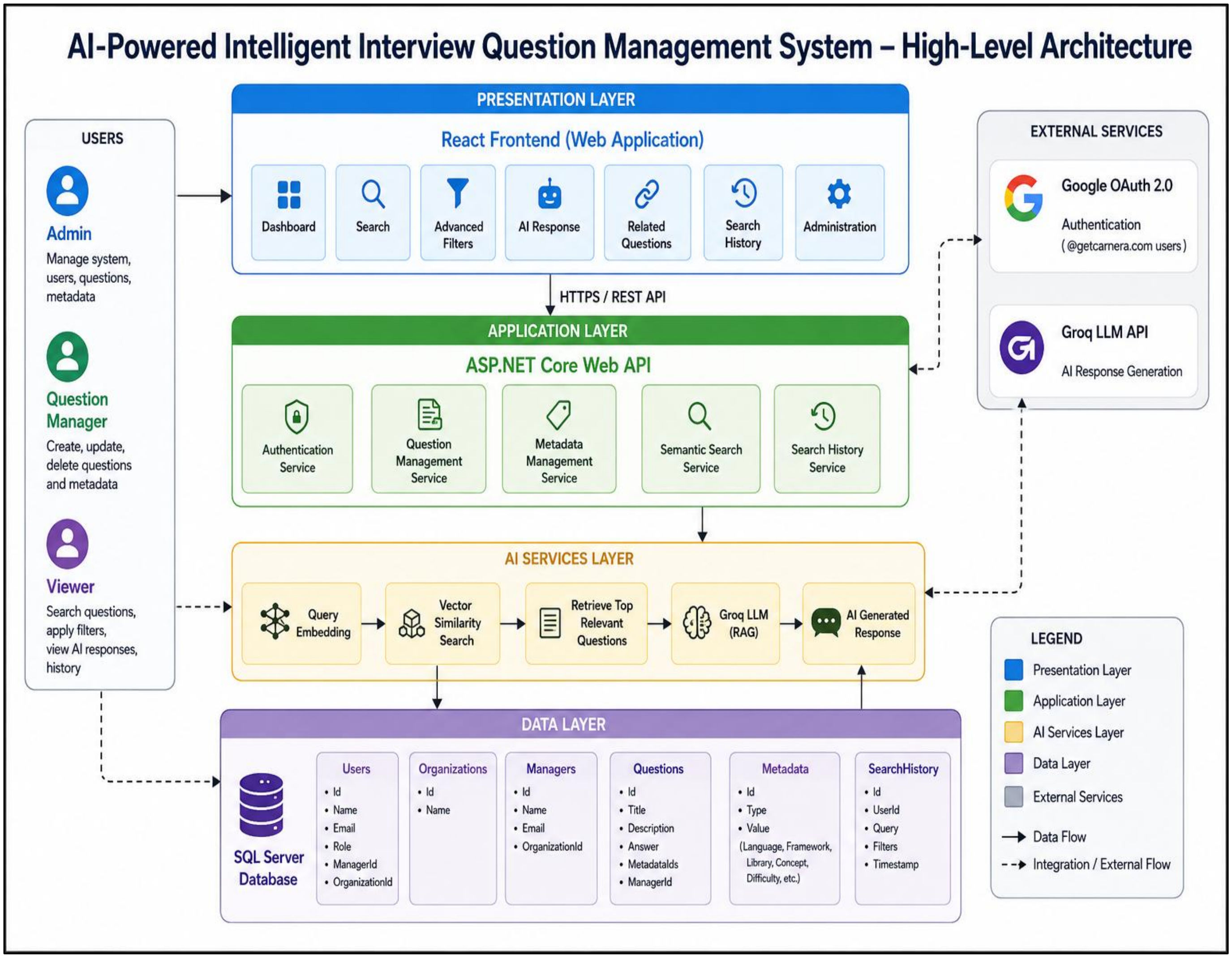
The **AI-Powered Intelligent Interview Question Management System** is a web-based application designed to centralize the storage, management, and retrieval of interview questions. The system enables Admins and Question Managers to maintain interview questions and their associated metadata, while authorized Viewers can search questions using natural language queries, apply advanced filters, and receive AI-generated responses powered by a Retrieval-Augmented Generation (RAG) pipeline.

The application follows a layered architecture where a React frontend communicates with an ASP.NET Core Web API backend. The backend manages authentication, business logic, metadata management, AI-powered search, and database operations to provide a secure and efficient interview question management platform.

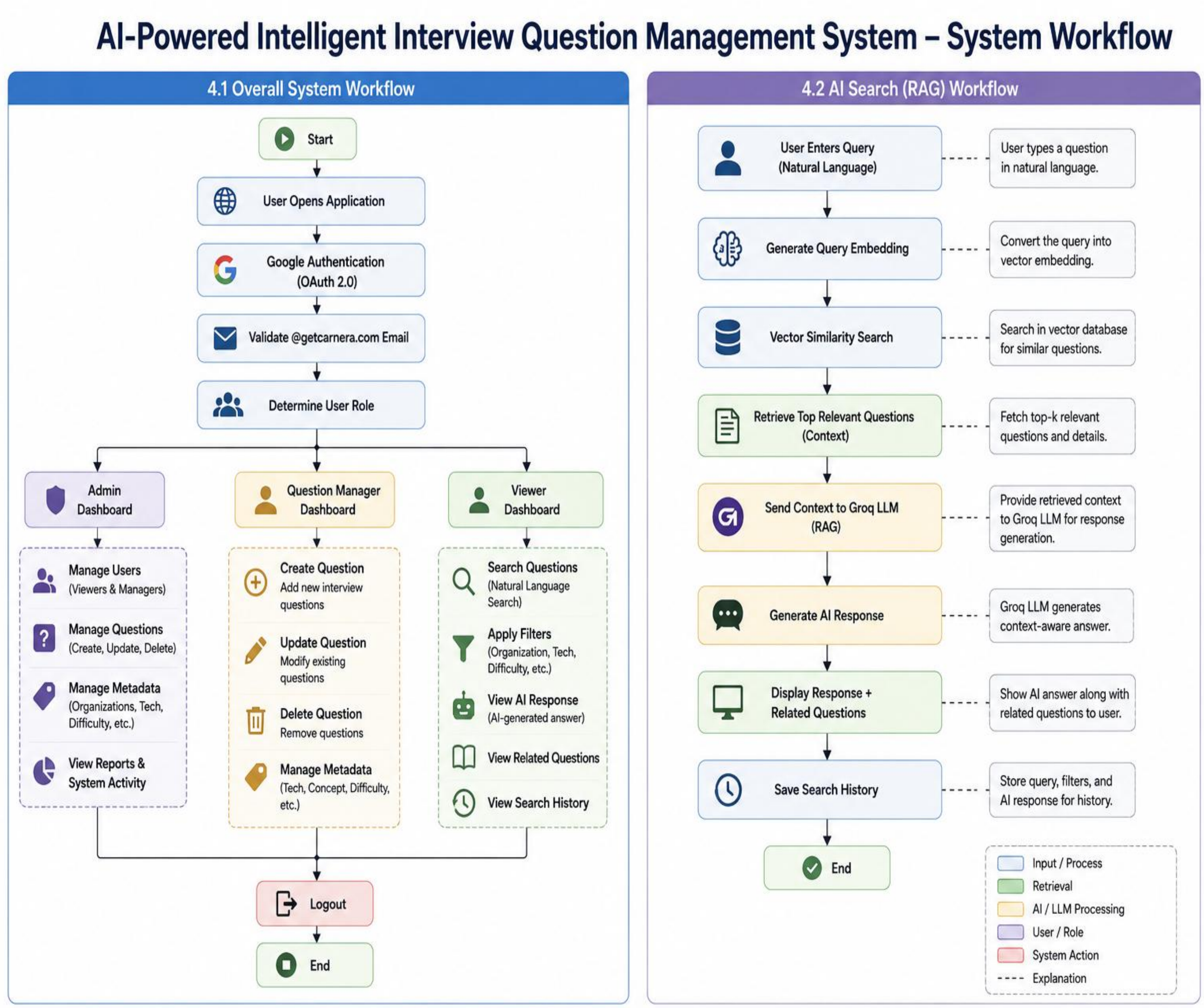
Technology Stack

Layer	**Technology**
Frontend	React.js
Backend	ASP.NET Core Web API (.NET 10)
Database	SQL Server
Authentication	Google OAuth 2.0
AI Model	Groq LLM
Embedding Model	Ollama (nomic-embed-text)
Vector Search	SQL Server Vector Search
API Testing	Swagger UI
Version Control	Git & GitHub
IDE	Visual Studio 2022 & Visual Studio Code

3. High-Level Architecture

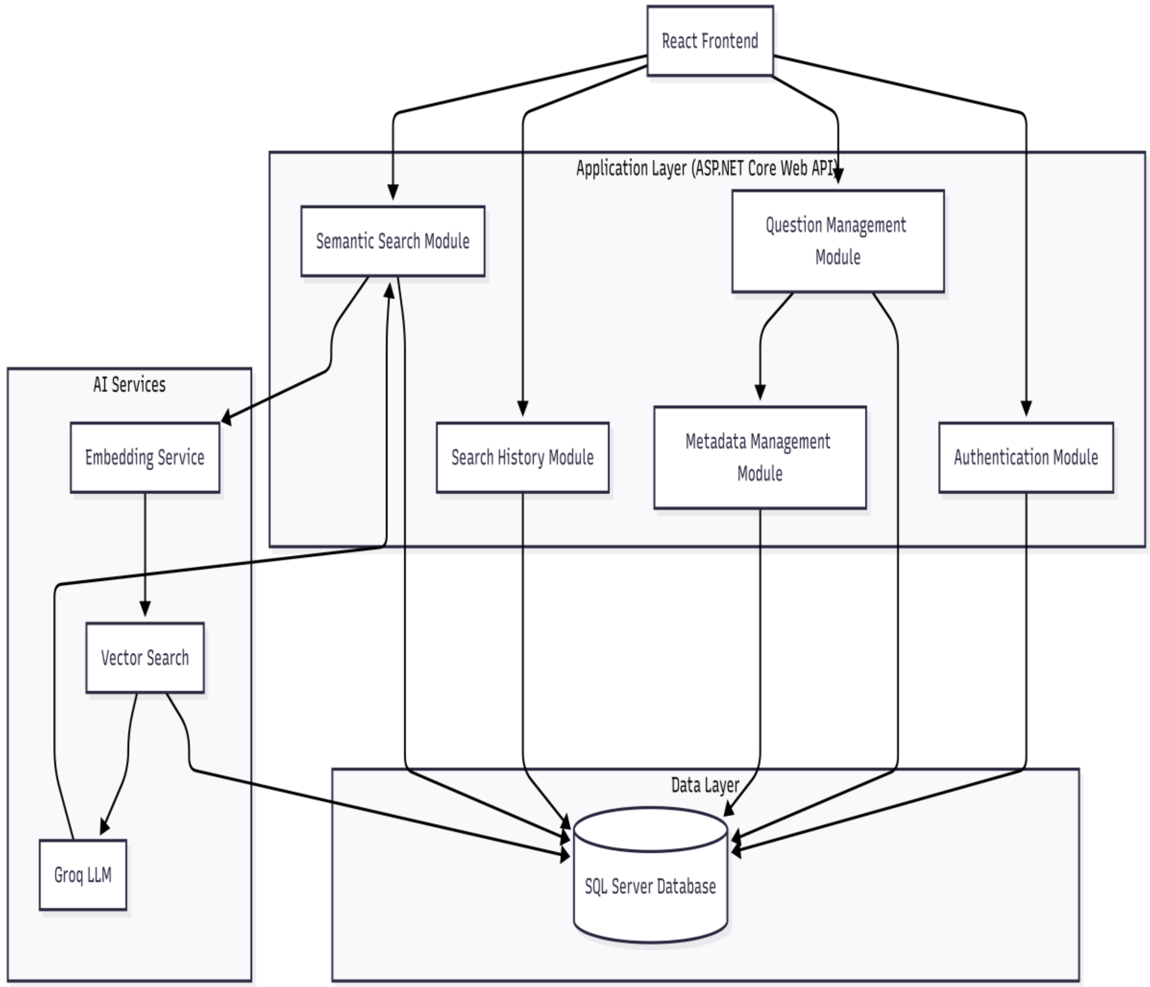


4. System Workflow



5. Module Design

Module Interaction Diagram



5.1 Authentication Module

Purpose

The Authentication Module verifies user identity using Google OAuth 2.0 and restricts access to users with a valid @getcarnera.com email address. After successful authentication, it determines the user's role (Admin, Question Manager, or Viewer) and grants appropriate permissions using Role-Based Access Control (RBAC).

Responsibilities

- Authenticate users via Google OAuth
- Validate corporate email domain
- Authorize access based on RBAC

5.2 Question Management Module

Purpose

This module allows Admins and Question Managers to create, update, delete, and organize interview questions while maintaining their associated metadata.

Responsibilities

- Create interview questions
- Update existing questions
- Delete questions
- Associate metadata with questions

5.3 Metadata Management Module

Purpose

This module manages classification data used for organizing and filtering interview questions.

Responsibilities

- Manage organizations
- Manage managers
- Manage programming languages
- Manage frameworks
- Manage libraries
- Manage concepts

- Manage difficulty levels

5.4 Semantic Search Module

Purpose

This module performs AI-powered semantic search using the Retrieval-Augmented Generation (RAG) pipeline. It converts natural language queries into embeddings, retrieves relevant questions through vector similarity search, and generates context-aware responses using the LLM.

Responsibilities

- Generate query embeddings
- Perform vector similarity search
- Retrieve relevant interview questions
- Generate AI-powered responses

5.5 Search History Module

Purpose

This module records user search activities, allowing users to review previous searches and revisit AI-generated responses.

Responsibilities

- Store search queries
- Store applied filters
- Display search history
- Retrieve previous searches

5.6 Viewer Management Module

Purpose

The Viewer Management Module enables Administrators to manage Viewer accounts and control their access to the system. It allows Administrators to create, update, and delete Viewer accounts, as well as assign Viewers to organizations, ensuring that authorized users have appropriate access based on organizational requirements.

Responsibilities

- Create Viewer accounts
- Update Viewer information
- Delete Viewer accounts
- Assign Viewers to organizations
- Manage Viewer access permissions

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6. API Design

API Overview

The AI-Powered Intelligent Interview Question Management System exposes RESTful APIs to facilitate communication between the React frontend and the ASP.NET Core Web API. These APIs enable user authentication, question management, metadata management, semantic search, and search history operations. Access to APIs is controlled using Role-Based Access Control (RBAC).

Proposed Endpoints

Note: The API endpoints presented in this section represent the proposed REST API design and may be refined during implementation.

Module	Endpoint	Method	Description	Access
Authentication	/api/auth/login	POST	Authenticate user using Google OAuth	All Users
Questions	/api/questions	GET	Retrieve all interview questions	All Users
Questions	/api/questions/{id}	GET	Retrieve a question by ID	All Users

Questions	/api/questions	POST	Create a new interview question	Admin, Question Manager
Questions	/api/questions/{id}	PUT	Update an existing question	Admin, Question Manager
Questions	/api/questions/{id}	DELETE	Delete an interview question	Admin, Question Manager
Search	/api/questions/search	GET	Perform AI-powered semantic search	All Users
Metadata	/api/metadata	GET	Retrieve metadata	All Users
Metadata	/api/metadata	POST	Create or update metadata	Admin, Question Manager
Search History	/api/search-history	GET	Retrieve user search history	All Users

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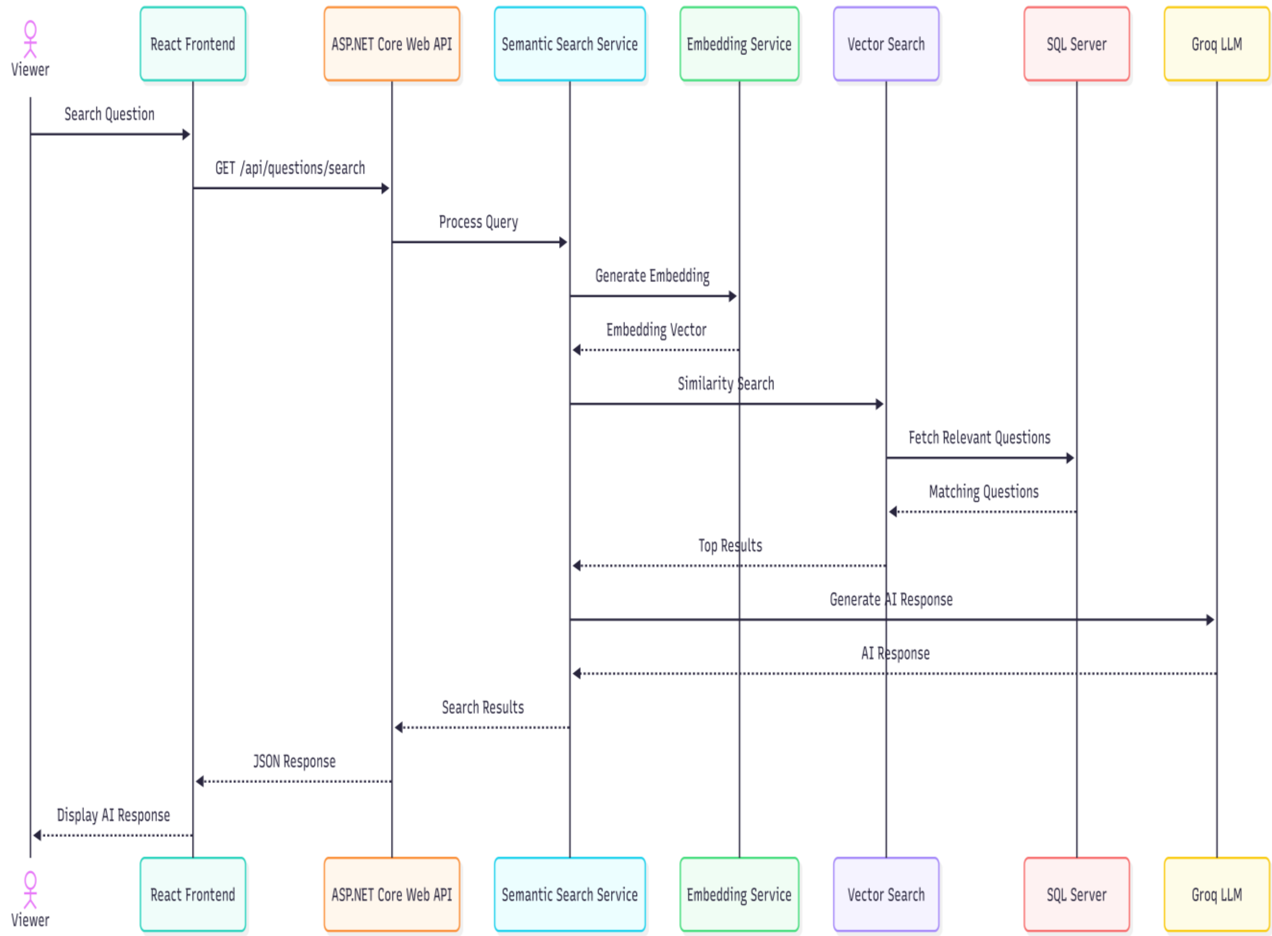
API Request Flow



Standard Success Response

```
{
  "success": true,
  "message": "Request processed successfully.",
  "data": { }
}
```

Standard Error Response

```
{
  "success": false,
  "message": "Unauthorized access.",
  "errors": [ ]
}
```